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3. Implementation

3.1. System Architecture

The system is built as a multi-stage pipeline designed to bridge the gap between unstructured job metadata and high-sparsity user clickstream data. It follows a "Two-Tower" logic where content features and behavioral features are processed separately before being fused in a hybrid ranking layer.

Data Ingestion Layer: Loads event_data.csv (clicks/purchases) and job metadata. It handles data type casting / IDs and filters to remove

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right parameters required experimentation and analysis. Plotting user and movie in latent space showed clear patterns, which suggests that SVD captures meaningful structure.

12. Conclusion

In conclusion, this assignment shows that latent factor models are much more effective than simple baseline methods for collaborative filtering on large datasets. Among all methods tested, SVD produced the best overall performance.

Using Maximum Likelihood Estimation also helped improve results by handling missing data more effectively. Overall, the assignment highlights the importance of understanding the data and choosing appropriate models rather than relying on overly simple or overly complex methods.



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